College of Technology Galle

Course: Full Stack Software Development — Object Oriented Programming — Module K72T004M03 — NVQ Level 05

ASSIGNMENT 02 — MARKING SCHEME

Lessons Covered: (01) Hello World, JDK & Java Program Anatomy | (02) Variables, Data Types & Console I/O

FOR EXAMINERS ONLY — DO NOT DISTRIBUTE TO STUDENTS

Duration: 1 Hour **Total Marks:** 40

Pass guidance: Award marks for equivalent correct answers; spelling of keywords may vary if meaning is clear.

Part	Questions	Marks
A — JDK, Tools & Program Anatomy	Q1–Q2	12
B — Variables, Data Types & Console I/O	Q3–Q4	14
C — Code Writing & Tracing	Q5–Q7	14
Total		**40**

Part A — Short Answers

[12 Marks]

Question 1 — (6 marks)

Part	Expected answer	Mark
1(a)	**Answer:** JDK = **Java Development Kit**. Required because it provides the tools to **compile** (`javac`) and **run** Java programs (and related libraries / JVM tools). *1 mark for expansion; 1 mark for a valid reason (compile/run/develop Java programs).*	2
1(b)	**Answer:** IDE = **VS Code** (Visual Studio Code). Extension pack = **Extension Pack for Java** (by Microsoft). *1 mark for VS Code; 1 mark for Extension Pack for Java (accept "Java Extension Pack").*	2
1(c)	**Answer:** Compile: `javac Hello.java` Run: `java Hello` *1 mark each. Do not accept `java Hello.java` for running.*	2

Question 2 — (6 marks)

Part	Expected answer	Mark

2(a)	**Answer:** `public` / `main` / `println`
	`public class Welcome`
	`{`
	`  public static void main(String[] args)`
	`  {`
	`  System.out.println("Hello, World!");`
	`  }`
	`}`
	1 mark per blank (3 marks).
		3

2(b)	**Answer:** The `main` method is the **entry point** / starting point of a Java application (where execution begins).	1
2(c)	**Answer:** `//` starts a **comment** (ignored by the compiler). `;` ends a **statement**. *1 mark for comment; 1 mark for statement terminator.*	2

Part B — Variables, Data Types & Console I/O

[14 Marks]

Question 3 — (5 marks)

Award **1 mark per row** if both a suitable type and a valid declaration + initialization (as two steps or combined) are shown. Accept equivalent names/values.

#	Type	Example (declaration + initialization)	Mark

1	`int`	`int age;`
		`age = 20;`
		or `int age = 20;`
			1

2	`double`	`double fee;` `fee = 15000.50;`	1
3	`String`	`String name;` `name = "Kasun";`	1
4	`boolean`	`boolean hasPaid;` `hasPaid = true;`	1
5	`char`	`char grade;` `grade = 'A';`	1

If type is wrong but declaration uses that wrong type consistently, award 0 for that row. Accept `float` for fee only if justified; preferred answer is `double` (as taught).

Question 4 — (9 marks)

Part	Expected answer	Mark
4(a)	**Answer:** `String` uses **double quotes** e.g. `"Hello"`. `char` uses **single quotes** e.g. `'A'`. *1 mark quotes + examples for String; 1 mark for char.*	2
4(b)	**Answer (any equivalent):** `Calculator calc = new Calculator();` or `Calculator obj = new Calculator();` *1 mark for type + name; 1 mark for `new Calculator()`. Accept any valid object name.*	2

4(c)	**Answer:**
	`Scanner sc = new Scanner(System.in);`
	`String line = sc.nextLine();`
	1.5 marks (or 2) for creating Scanner with `System.in`; 1.5 marks (or 1) for `nextLine()`. Total 3. Award full marks if `import java.util.Scanner;` is also written but not required unless asked. Accept other variable names.
		3

4(d)	**Answer:** `System.out.println("Hello " + name + ", you are " + age);` *Accept minor spacing differences if output matches. 1 mark concatenation structure; 1 mark correct text/punctuation.*	2

Part C — Code Writing & Tracing

[14 Marks]

Question 5 — Trace the output (4 marks)

Expected console output:

```
Pen x 3
Total = 76.5
```

Award	Mark
First line correct: `Pen x 3`	2
Second line correct: `Total = 76.5` (accept `76.50`)	2

If calculation is wrong but format of both lines is right, award max 2 marks total.

Question 6 — Find and correct errors (4 marks)

Award **1 mark per error** only if the student both identifies the fault and gives a correct fix.

#	Error	Corrected line / fix	Mark
1	`Main` should be `main` (case-sensitive)	`public static void main(String[] args)`	1
2	`String args` should be `String[] args` (array)	`String[] args`	1
3	`string` / single quotes wrong for text — use `String` and double quotes	`String name = "Nimal";`	1
4	Missing semicolon at end of `println` statement	`System.out.println("Hi " + name);`	1

Note: `string` (lowercase) and `'Nimal'` may be counted as one combined error or two; if student lists both clearly with fixes, still cap this item at 1 mark for the name line, and ensure four distinct marks total. Prefer the four rows above.

Question 7 — Write a complete program (6 marks)

Model answer (equivalent accepted):

```
import java.util.Scanner;

public class StudentCard {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String name = sc.nextLine();
        int mark = Integer.parseInt(sc.nextLine());
        // accept: int mark = sc.nextInt();
        System.out.println("Student: " + name + " | Mark: " + mark);
    }
}
```

Criteria	Mark
Correct class name `StudentCard` and valid `main` method structure	1
Required `import java.util.Scanner;` (or fully-qualified use of Scanner)	1
Creates `Scanner` reading from `System.in`	1
Reads name into a `String` variable	1
Reads mark into an `int` (accept `nextInt()` or `parseInt(nextLine())`)	1
Prints in the required form: `Student: ... | Mark: ...`	1

Do not heavily penalize missing braces style or indentation if the program is logically correct. Package line is optional.

******* END OF MARKING SCHEME *******